

LINEAR STATISTICAL MODELS — MID-SEM SHEET

[24] = in 2024 paper [25] = in 2025 paper [G] = in the professor's guide

1. PROJECTIONS & IDEMPOTENT MATRICES [24 Q1, 25 Q4]

TFAE (projection). $\mathbf{P}$ proj. matrix $\iff \mathbf{P}^2 = \mathbf{P} \iff \mathcal{N}(\mathbf{P}) = \mathcal{C}(I - \mathbf{P}) \iff \rho(\mathbf{P}) + \rho(I - \mathbf{P}) = n$.

TFAE (orthogonal projector). $\mathbf{P}$ orth. proj. $\iff \mathbf{P} = \mathbf{P}^T \mathbf{P} \iff [\mathbf{P}^T = \mathbf{P} \text{ and } \mathbf{P}^2 = \mathbf{P}]$.

Consequences. eigenvalues of idempotent $\in \{0, 1\}$; hence

$$\text{tr}(\mathbf{P}) = \rho(\mathbf{P})$$

$\mathcal{C}(\mathbf{P}) = S$, $\mathcal{C}(I - \mathbf{P}) = S^\perp$, $\mathbf{P}$ nnd, $\mathbf{Y}^T \mathbf{P} \mathbf{Y} = \|\mathbf{P} \mathbf{Y}\|^2 \geq 0$.

Formula. $\mathbf{P}_\mathbf{X} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$, invariant to the g-inverse. $\mathbf{P}_\mathbf{X} \mathbf{X} = \mathbf{X}$, $(I - \mathbf{P}_\mathbf{X}) \mathbf{X} = 0$, $\rho(\mathbf{P}_\mathbf{X}) = \rho(\mathbf{X})$. If $\mathcal{C}(\mathbf{Z}) \subseteq \mathcal{C}(\mathbf{X})$ then $\mathbf{P}_\mathbf{X} \mathbf{P}_\mathbf{Z} = \mathbf{P}_\mathbf{Z} \mathbf{P}_\mathbf{X} = \mathbf{P}_\mathbf{Z}$.
Decomposition lemma. $\mathbf{X} = [\mathbf{Z} : \mathbf{W}] \Rightarrow$

$$\mathbf{P}_\mathbf{X} = \mathbf{P}_\mathbf{Z} + \mathbf{P}_{\mathbf{W}|\mathbf{Z}}$$

$\mathbf{P}_{\mathbf{W}|\mathbf{Z}}$ = orth. proj. onto $\mathcal{C}((I - \mathbf{P}_\mathbf{Z})\mathbf{W})$; explicitly $(I - \mathbf{P}_\mathbf{Z})\mathbf{W}(\mathbf{W}^T(I - \mathbf{P}_\mathbf{Z})\mathbf{W})^{-1} \mathbf{W}^T(I - \mathbf{P}_\mathbf{Z})$.

[24 Q1] $\mathbf{P}, \mathbf{Q}$ symm. orth. proj.: $\mathbf{P} + \mathbf{Q}$ is an orth. proj. $\iff \mathbf{PQ} = \mathbf{O}$; then $\rho(\mathbf{P}) + \rho(\mathbf{Q}) = \rho(\mathbf{P} + \mathbf{Q})$. ($\Leftarrow$) $\mathbf{PQ} = \mathbf{O} \Rightarrow \mathbf{QP} = (\mathbf{PQ})^T = \mathbf{O}$; then $(\mathbf{P} + \mathbf{Q})^2 = \mathbf{P}^2 + \mathbf{PQ} + \mathbf{QP} + \mathbf{Q}^2 = \mathbf{P} + \mathbf{Q}$, and it is symmetric.

($\Rightarrow$) idempotence gives $\mathbf{PQ} + \mathbf{QP} = \mathbf{O}$. Multiply on the left by $\mathbf{P}$: $\mathbf{PQ} + \mathbf{PQP} = \mathbf{O}$; on the right by $\mathbf{P}$: $\mathbf{PQP} + \mathbf{QP} = \mathbf{O}$. Subtract $\Rightarrow \mathbf{PQ} = \mathbf{QP}$; combined with the sum $= \mathbf{O}$ gives $2\mathbf{PQ} = \mathbf{O}$.

Rank: $\rho(\mathbf{P} + \mathbf{Q}) = \text{tr}(\mathbf{P} + \mathbf{Q}) = \text{tr} \mathbf{P} + \text{tr} \mathbf{Q} = \rho(\mathbf{P}) + \rho(\mathbf{Q})$.

[25 Q4] $\mathbf{A}, \mathbf{B}$ symm., $\mathbf{B}$ idempotent, $\mathbf{A} \succeq \mathbf{0}$, $I - \mathbf{A} - \mathbf{B} \succeq \mathbf{0} \Rightarrow \mathbf{AB} = \mathbf{BA} = \mathbf{O}$. Let $y \in \mathcal{C}(\mathbf{B})$, so $\mathbf{B}y = y$ ($\mathbf{B}$ is the orth. projector onto $\mathcal{C}(\mathbf{B})$). Then $0 \leq y^T(I - \mathbf{A} - \mathbf{B})y = y^T y - y^T \mathbf{A} y - y^T y = -y^T \mathbf{A} y$, so $y^T \mathbf{A} y \leq 0$; with $\mathbf{A} \succeq \mathbf{0}$, $y^T \mathbf{A} y = 0$. Write $\mathbf{A} = L^T L$: $\|Ly\|^2 = 0 \Rightarrow Ly = 0 \Rightarrow \mathbf{A}y = 0$. True for every column of $\mathbf{B}$, so $\mathbf{AB} = \mathbf{O}$; transpose gives $\mathbf{BA} = \mathbf{O}$.

2. GENERALISED INVERSES [24 Q3]

G is a g-inverse of A iff

$$AGA = A$$

. If $y \in \mathcal{C}(A)$ then $x = Gy$ solves $Ax = y$; **full solution set** = $A^-y + (I - A^-A)z$, z arbitrary.

Symmetric $A = P\Delta P^T$. $A^- = P\Delta P^T$ with $\delta_i = 1/\lambda_i$ if $\lambda_i \neq 0$, arbitrary if $\lambda_i = 0$. Moore–Penrose A^+ : take $\delta_i = 0$ there; gives the min-norm solution.

Facts. $A^T(AA^T)^-$ and $(A^T A)^-A^T$ are g-inverses of A . AG idempotent with $\rho(AG) = \rho(A) \iff G = A^-$. **Invariance:** if $\mathcal{R}(A) \subseteq \mathcal{R}(B)$ and $\mathcal{C}(C) \subseteq \mathcal{C}(B)$ then AB^-C does not depend on B^- . Reflexive: $A^-AA^- = A^-$.

[24 Q3] Block g-inverse. $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, A_{11} is $r \times r$, $\rho(A) = \rho(A_{11}) = r$.

Then $G = \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & 0 \end{bmatrix}$ is a g-inverse.

Proof: $AG = \begin{bmatrix} I_r & 0 \\ A_{21}A_{11}^{-1} & 0 \end{bmatrix}$, so $AGA = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{21}A_{11}^{-1}A_{12} \end{bmatrix}$. Since $\rho(A) = \rho(A_{11}) = r$, the last $m - r$ rows of A are combinations of the first r : $[A_{21} \ A_{22}] = M[A_{11} \ A_{12}]$. Then $A_{21} = MA_{11} \Rightarrow M = A_{21}A_{11}^{-1}$, hence $A_{22} = MA_{12} = A_{21}A_{11}^{-1}A_{12}$. So $AGA = A$.

3. ESTIMABILITY & GAUSS–MARKOV [24 Q2, 25 Q2 — BOTH YEARS]

Model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\text{Var}\boldsymbol{\varepsilon} = \sigma^2 I$. **Normal eq.** $\mathbf{X}^T \mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T \mathbf{Y}$ — always solvable ($\mathbf{X}^T \mathbf{Y} \in \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T \mathbf{X})$); unique iff $\rho(\mathbf{X}) = p$.

Identifiable $\iff \mathbf{X}^T \mathbf{X}$ non-singular $\iff \rho(\mathbf{X}) = p$.

Estimable = an unbiased estimator exists.

$$a^T \boldsymbol{\beta} \text{ estimable} \iff a \in \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T \mathbf{X})$$

Why: $\mathbb{E}(\ell^T \mathbf{Y}) = a^T \boldsymbol{\beta} \forall \boldsymbol{\beta} \iff \ell^T \mathbf{X} = a^T \iff a = \mathbf{X}^T \ell$.

FAST TEST (use this). Find a basis $\{z\}$ of $\mathcal{N}(\mathbf{X})$ once. Then

$$a^T \boldsymbol{\beta} \text{ estimable} \iff a^T z = 0 \quad \forall z \in \mathcal{N}(\mathbf{X})$$

because $\mathcal{C}(\mathbf{X}^T) = \mathcal{N}(\mathbf{X})^\perp$.

Uniqueness. If $a^T \boldsymbol{\beta}$ estimable, $a^T \hat{\boldsymbol{\beta}}$ is the same for every solution $\hat{\boldsymbol{\beta}}$ ($a = \mathbf{X}^T \mathbf{X}c \Rightarrow a^T \hat{\boldsymbol{\beta}} = c^T \mathbf{X}^T \mathbf{Y}$).

LZF. $c^T \mathbf{Y}$ is a linear zero function iff $\mathbb{E}(c^T \mathbf{Y}) = 0 \forall \boldsymbol{\beta}$ iff $\mathbf{X}^T c = 0$ (i.e. $c \perp \mathcal{C}(\mathbf{X})$).

BLUE criterion. LUE $\ell^T \mathbf{Y}$ is BLUE $\iff \text{Cov}(\ell^T \mathbf{Y}, c^T \mathbf{Y}) = 0$ for every LZF $c^T \mathbf{Y}$.

GAUSS–MARKOV THM. If $a^T \boldsymbol{\beta}$ is estimable then $a^T \hat{\boldsymbol{\beta}}$ is its unique BLUE, $\hat{\boldsymbol{\beta}}$ any solution of the normal equations. $\text{Var}(a^T \hat{\boldsymbol{\beta}}) = \sigma^2 a^T (\mathbf{X}^T \mathbf{X})^- a$. **Proof:** $a = \mathbf{X}^T \mathbf{X}u \Rightarrow a^T \hat{\boldsymbol{\beta}} = u^T \mathbf{X}^T \mathbf{Y}$; for LZF $h^T \mathbf{Y}$ ($\mathbf{X}^T h = 0$), $\text{Cov} = \sigma^2 u^T \mathbf{X}^T h = 0$. ✓

$\hat{\boldsymbol{\mu}}$ and $\mathbf{X}^T \boldsymbol{\mu}$. $\hat{\boldsymbol{\mu}} = \mathbf{P}_\mathbf{X} \mathbf{Y}$. $\theta = \mathbf{X}^T \boldsymbol{\mu} = \mathbf{X}^T \mathbf{X}\boldsymbol{\beta}$ is always estimable, BLUE = $\mathbf{X}^T \mathbf{Y}$ (since $\mathbf{X}^T \mathbf{P}_\mathbf{X} = \mathbf{X}^T$). **[25 Q2b]**

Estimate of σ^2 . $\hat{\sigma}^2 = \text{MSE} = \frac{\mathbf{Y}^T(I - \mathbf{P}_\mathbf{X})\mathbf{Y}}{n - \rho(\mathbf{X})}$, unbiased. **Use $n - \rho(\mathbf{X})$,**

the RANK.

GLS. If $\text{Var}(\boldsymbol{\varepsilon}) = \Sigma \succ 0$: BLUE from $\mathbf{X}^T \Sigma^{-1} \mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T \Sigma^{-1} \mathbf{Y}$; $\text{Var}(a^T \hat{\boldsymbol{\beta}}^{GLS}) = a^T (\mathbf{X}^T \Sigma^{-1} \mathbf{X})^{-1} a \leq \text{Var}(a^T \hat{\boldsymbol{\beta}}^{LS}) =$

$$a^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \Sigma \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} a.$$

4. TWO WORKED DESIGNS (memorise the method)

[25 Q2] $Y_{ijk} = \alpha_i + \beta_j + \varepsilon_{ijk}$, $i, j \in \{1, 2\}$, $k \leq m$. $\boldsymbol{\beta} = (\alpha_1, \alpha_2, \beta_1, \beta_2)^T$; 4 distinct rows $(1, 0, 1, 0), (1, 0, 0, 1), (0, 1, 1, 0), (0, 1, 0, 1)$. $\rho(\mathbf{X}) = 3$ ($\text{col}_1 + \text{col}_2 = \text{col}_3 + \text{col}_4$). $\mathcal{N}(\mathbf{X}) = \text{span}\{z\}$,

$$z = (1, 1, -1, -1)^T$$

Test: $a_1 + a_2 - a_3 - a_4 = 0$. Fitted cell means $\hat{\mu}_{ij} = \bar{Y}_{i..} + \bar{Y}_{.j.} - \bar{Y}...$

$a^T \boldsymbol{\beta}$	$a^T z$	Est.?	BLUE (Var)
α_1	1	No	—
$\alpha_1 - \alpha_2$	0	Yes	$\bar{Y}_{1..} - \bar{Y}_{2..}$ (σ^2/m)
$\alpha_1 + \alpha_2 + \beta_1 + \beta_2$	0	Yes	$2\bar{Y}...$ (σ^2/m)
$\alpha_1 - \alpha_2 + \beta_1 - \beta_2$	0	Yes	$(\bar{Y}_{1..} - \bar{Y}_{2..}) + (\bar{Y}_{.1.} - \bar{Y}_{.2.})$ ($2\sigma^2/m$)

(3rd row = $\mu_{11} + \mu_{22}$; 4th = $\mu_{11} - \mu_{22}$.)

[24 Q2] $Y_{1j} = \beta_1$, $Y_{2j} = \beta_1 + \beta_2$, $Y_{3j} = \beta_2 + \beta_3$, $Y_{4j} = \beta_3 + \beta_4$. Dis-

tinct rows $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$, $\det = 1 \Rightarrow$ **full rank 4: EVERYTHING estimable**

(a trap — always check the rank first). $\hat{\beta}_1 = \bar{Y}_{1..}$, $\hat{\beta}_2 = \bar{Y}_{2..} - \bar{Y}_{1..}$, $\hat{\beta}_3 = \bar{Y}_{3..} - \bar{Y}_{2..} + \bar{Y}_{1..}$, $\hat{\beta}_4 = \bar{Y}_{4..} - \bar{Y}_{3..} + \bar{Y}_{2..} - \bar{Y}_{1..}$; $\text{Var} = \frac{\sigma^2}{m}, \frac{2\sigma^2}{m}, \frac{3\sigma^2}{m}, \frac{4\sigma^2}{m}$.

$H_0 : \beta_1 = \beta_2$: $\hat{\beta}_1 - \hat{\beta}_2 = 2\bar{Y}_{1..} - \bar{Y}_{2..}$, $\text{Var} = 5\sigma^2/m$, $T = \frac{2\bar{Y}_{1..} - \bar{Y}_{2..}}{\hat{\sigma}\sqrt{5/m}} \sim t_{4m-4}$,

$F = T^2 \sim F_{1, 4m-4}$.

5. NORMAL THEORY & QUADRATIC FORMS

$\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$, A symmetric:

- $\mathbb{E}(\mathbf{Y}^T A \mathbf{Y}) = \text{tr}(A\Sigma) + \boldsymbol{\mu}^T A \boldsymbol{\mu}$
- $\text{Cov}(\mathbf{Y}, \mathbf{Y}^T A \mathbf{Y}) = 2\Sigma A \boldsymbol{\mu}$
- $\text{Var}(\mathbf{Y}^T A \mathbf{Y}) = 2\text{tr}((A\Sigma)^2) + 4\boldsymbol{\mu}^T A \Sigma A \boldsymbol{\mu}$
- $\mathbf{Y}^T A \mathbf{Y} \perp\!\!\!\perp B \mathbf{Y} \iff B \Sigma A = 0$
- $A \mathbf{Y} \perp\!\!\!\perp B \mathbf{Y} \iff A \Sigma B^T = 0$

KEY THEOREM. $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, I)$, A symm.:

$$\mathbf{Y}^T A \mathbf{Y} \sim \chi_{\rho(A)}^2 \left(\frac{1}{2} \boldsymbol{\mu}^T A \boldsymbol{\mu} \right) \iff A^2 = A$$

Central ($\lambda = 0$) iff $A\boldsymbol{\mu} = 0$. MGF of χ_n^2 : $(1 - 2t)^{-n/2}$, $t < \frac{1}{2}$. Proof: $|\det(I - 2tA)|^{-1/2} = \prod (1 - 2t\lambda_j)^{-1/2} = (1 - 2t)^{-r/2} \forall t \iff \lambda_j = 1$.

Cor. $\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I)$, $A = I - \mathbf{P}_\mathbf{X}$: $\text{SSE}/\sigma^2 \sim \chi_{n-\rho(\mathbf{X})}^2$, and $\text{SSE} \perp\!\!\!\perp \hat{\boldsymbol{\beta}}$ (since $(I - \mathbf{P}_\mathbf{X})\mathbf{X} = 0$).

FISHER–COCHRAN. $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, I)$, A_j symm. nnd, $\mathbf{Y}^T \mathbf{Y} = \sum_j \mathbf{Y}^T A_j \mathbf{Y}$. Then each $\mathbf{Y}^T A_j \mathbf{Y} \sim \chi_{\rho(A_j)}^2 (\lambda_j/2)$ and they are mutually independent

$$\iff \sum_j \rho(A_j) = n$$

in which case $\lambda_j = \boldsymbol{\mu}^T A_j \boldsymbol{\mu}$, $\sum \lambda_j = \boldsymbol{\mu}^T \boldsymbol{\mu}$, each A_j idempotent, $A_j A_k = 0$. **This is what licenses the ANOVA table: the d.f. column adding up IS the hypothesis.**

6. TESTING: WALD = F = LRT [25 Q1, Q3]

$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$, $\rho(\mathbf{X}) = p$. $H_0 : L^T \boldsymbol{\beta} = \theta_0$, $\rho(L) = r$.

Constrained LSE.

$$\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} - (\mathbf{X}^T \mathbf{X})^{-1} L (L^T (\mathbf{X}^T \mathbf{X})^{-1} L)^{-1} (\hat{\theta} - \theta_0)$$

$$\hat{\theta} = L^T \hat{\boldsymbol{\beta}} \sim N_r(L^T \boldsymbol{\beta}, \sigma^2 L^T (\mathbf{X}^T \mathbf{X})^{-1} L).$$

Reduction in SSE (the bridge).

$$\text{SSE}_{red} - \text{SSE}_{full} = (\hat{\theta} - \theta_0)^T (L^T (\mathbf{X}^T \mathbf{X})^{-1} L)^{-1} (\hat{\theta} - \theta_0)$$

Wald T^2 . $W = \frac{1}{\sigma^2} (\hat{\theta} - \theta_0)^T (L^T (\mathbf{X}^T \mathbf{X})^{-1} L)^{-1} (\hat{\theta} - \theta_0)$,

$$\frac{W}{r} \sim F_{r, n-p}(\lambda/2)$$

$\lambda = \sigma^{-2} (L^T \boldsymbol{\beta} - \theta_0)^T (L^T (\mathbf{X}^T \mathbf{X})^{-1} L)^{-1} (L^T \boldsymbol{\beta} - \theta_0)$. Under H_0 : $\lambda = 0$. Power $\uparrow$ as $\lambda \uparrow \infty$. $W \sim r F_{r, n-p}$ — **don't lose the r .**

F-test.

$$F = \frac{(\text{SSE}_{red} - \text{SSE}_{full}) / (\text{df}_{red} - \text{df}_{full})}{\text{MSE}_{full}} = \frac{W}{r}$$

$\text{df}_{full} = n - p$, $\text{df}_{red} = n - (p - r)$. Reject if $F > F_{r, n-p}^{1-\alpha}$.

LRT. $\Lambda = \left(\frac{\text{SSE}_{full}}{\text{SSE}_{red}} \right)^{n/2}$; reject for small $\Lambda \iff$ large $\frac{\text{SSE}_{red} - \text{SSE}_{full}}{\text{SSE}_{full}}$

$\iff$ large F . **All three tests are the same rejection region.** $-2 \log \Lambda \xrightarrow{d} \chi_r^2$.

Sub-model form. $\mathbf{X} = [\mathbf{Z} : \mathbf{W}]$, $H_0 : \boldsymbol{\delta} = \mathbf{0}$: $F = \frac{\mathbf{Y}^T \mathbf{P}_{\mathbf{W}|\mathbf{Z}} \mathbf{Y} / s}{\mathbf{Y}^T (\mathbf{I} - \mathbf{P}_{\mathbf{X}}) \mathbf{Y} / (n - p)}$,

with $\mathbf{Y}^T \mathbf{Y} = \mathbf{Y}^T \mathbf{P}_{\mathbf{Z}} \mathbf{Y} + \mathbf{Y}^T \mathbf{P}_{\mathbf{W}|\mathbf{Z}} \mathbf{Y} + \mathbf{Y}^T (\mathbf{I} - \mathbf{P}_{\mathbf{X}}) \mathbf{Y}$, ranks $q + s + (n - p) = n$. **FUNDAMENTAL THEOREMS OF LS.** 1st: $\text{SSE} = \min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 \sim \sigma^2 \chi_{n-p}^2$.

2nd: $\mathcal{C}(L) \subseteq \mathcal{C}(\mathbf{X}^T)$; then SSE and $\text{SSE}_{red} - \text{SSE}$ are *independent*, and $\text{SSE}_{red} - \text{SSE} \sim \sigma^2 \chi_m^2(\delta)$ with $\delta = 0$ iff $L^T \boldsymbol{\beta} = \theta_0$.

3rd [25 Q1]: $\mathbf{X} = [\mathbf{Z} : \mathbf{W}]$, $\rho(\mathbf{X}) = p$, $\rho(\mathbf{Z}) = q < p$, $S_1 = \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$, $S_0 = \min_{\boldsymbol{\gamma} \in \mathbb{R}^q} \|\mathbf{Y} - \mathbf{Z}\boldsymbol{\gamma}\|^2$. If $\mathbb{E}\mathbf{Y} \in \mathcal{C}(\mathbf{Z})$:

$$\frac{S_1}{S_0} \sim \text{Beta}\left(\frac{n-p}{2}, \frac{p-q}{2}\right)$$

Proof: $S_1 \sim \sigma^2 \chi_{n-p}^2$, $S_0 - S_1 \sim \sigma^2 \chi_{p-q}^2$ (central, as H_0 true), independent by the 2nd thm; and $\frac{A}{A+B} \sim \text{Beta}\left(\frac{a}{2}, \frac{b}{2}\right)$ for $A \sim \chi_a^2 \perp B \sim \chi_b^2$. Here $\frac{S_1}{S_0} = \frac{S_1}{S_1 + (S_0 - S_1)}$.

7. PARTIALLING OUT / COLLINEARITY [24 Q4, 25 Q3]

Frisch-Waugh. $\hat{\boldsymbol{\delta}} = (\mathbf{W}^T (\mathbf{I} - \mathbf{P}_{\mathbf{Z}}) \mathbf{W})^{-1} \mathbf{W}^T (\mathbf{I} - \mathbf{P}_{\mathbf{Z}}) \mathbf{Y}$ — i.e. regress $(\mathbf{I} - \mathbf{P}_{\mathbf{Z}}) \mathbf{Y}$ on $(\mathbf{I} - \mathbf{P}_{\mathbf{Z}}) \mathbf{W}$. “Effect of $\mathbf{W}$ adjusted for $\mathbf{Z}$.”

[24 Q4a] **Single coefficient.** $\mathbf{X}_{-j} = \mathbf{X}$ with column j removed, $d_j := \mathbf{X}_{-j}^T (\mathbf{I} - \mathbf{P}_{-j}) \mathbf{X}_{-j}$:

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\mathbf{X}_{-j}^T (\mathbf{I} - \mathbf{P}_{-j}) \mathbf{X}_{-j}} = \frac{\sigma^2}{d_j}$$

[24 Q4b] **t -statistic decomposition.** $\hat{\beta}_j \sqrt{d_j} / \sigma \sim N(\beta_j \sqrt{d_j} / \sigma, 1)$, so

$$T_j = \frac{\hat{\beta}_j}{\hat{\sigma} / \sqrt{d_j}} = \frac{1}{\hat{\sigma} / \sigma} \left(\frac{\beta_j}{\sigma} \sqrt{d_j} + W_j \right)$$

with $W_j \sim N(0, 1)$ independent of $\hat{\sigma}^2$.

[24 Q4c] **Collinearity moral.** If $\mathbf{X}_{-j_0}$ is nearly in $\mathcal{C}(\mathbf{X}_{-j_0})$ then $(\mathbf{I} - \mathbf{P}_{-j_0}) \mathbf{X}_{-j_0} \approx \mathbf{0}$, so $d_{j_0} \approx 0$, the non-centrality $\beta_{j_0} \sqrt{d_{j_0}} / \sigma \approx 0$, and $T_{j_0} \approx W_{j_0} / (\hat{\sigma} / \sigma)$ — *no power*. In the true one-regressor model $d = \|\mathbf{X}_{-j_0}\|^2$ is large, so the same β_{j_0} is highly significant. **Collinearity inflates $\text{Var}(\hat{\beta}_j)$ and destroys the t -test, without biasing anything.**

[25 Q3] **ANCOVA** $Y_{ij} = \mu_i + \beta_1 z_{ij} + \beta_2 w_{ij} + \varepsilon_{ij}$. $(\mathbf{I} - \mathbf{P}_{\mathbf{Z}}) =$ centring *within each treatment*; write $S_{zz} = \sum_i \sum_j (z_{ij} - \bar{z}_i)^2$ etc. Then

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} S_{zz} & S_{zw} \\ S_{zw} & S_{ww} \end{bmatrix}^{-1} \begin{bmatrix} S_{zy} \\ S_{wy} \end{bmatrix} =: M^{-1} \mathbf{s}$$

$\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 M^{-1}$, error d.f. = $n - r - 2$.

$H_0 : \beta_2 = 0$: $(M^{-1})_{22} = (S_{ww} - S_{zw}^2 / S_{zz})^{-1}$, so reject for large

$$U = \left(S_{ww} - \frac{S_{zw}^2}{S_{zz}} \right) \frac{\hat{\beta}_2^2}{\hat{\sigma}^2} \sim F_{1, n-r-2}$$

$H_0 : \beta_1 = \beta_2 = 0$: reject for large $V = \frac{1}{\hat{\sigma}^2} \hat{\boldsymbol{\beta}}^T M \hat{\boldsymbol{\beta}}$, $V/2 \sim F_{2, n-r-2}$.

8. SIMPLE LINEAR REGRESSION

$S_x := \sum (x_i - \bar{x})^2$ (**no division**). $\hat{\beta}_1 = S_{xy} / S_x$, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$; $\hat{\beta}_1 = \sum c_i Y_i$, $c_i = (x_i - \bar{x}) / S_x$.

$$\text{Var} \hat{\beta}_1 = \frac{\sigma^2}{S_x^2}, \quad \text{Var} \hat{\beta}_0 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_x^2} \right), \quad \text{Cov} = -\frac{\sigma^2 \bar{x}}{S_x^2}$$

Normal eqns $\equiv \sum e_i = 0$, $\sum e_i x_i = 0$; $\hat{\sigma}^2 = \text{SSE} / (n - 2)$; $\hat{\sigma}_{ML}^2 = \frac{n-2}{n} \hat{\sigma}^2$ (biased). $\text{SST} = \text{SSR} + \text{SSE}$; $R^2 = \text{SSR} / \text{SST} = \text{corr}(x, Y)^2$. $\hat{Y}_h = \bar{Y} + \hat{\beta}_1 (x_h - \bar{x})$ with $\bar{Y} \perp \hat{\beta}_1$, so

$$\text{Var}(\hat{Y}_h) = \sigma^2 \left(\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2} \right).$$

CI for mean $\hat{Y}_h \pm t_{n-2}^{1-\alpha/2} \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2}}$; **PI for new Y** same with

$$1 + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2}$$

inside. Ellipse CS: $(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T (\mathbf{X}^T \mathbf{X}) (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) \leq 2\hat{\sigma}^2 F_{2, n-2}^{1-\alpha}$. Bonferroni rectangle uses $t_{n-2}^{1-\alpha/4}$. Leave-one-out: $\hat{\beta}_1 = \hat{\beta}_1^{(-n)} + \frac{x_n - \bar{x}}{S_x^2} e_n^{(-n)}$. R^2 never decreases when a predictor is added ($\mathbf{Y}^T \mathbf{P}_{\mathbf{W}|\mathbf{Z}} \mathbf{Y} \geq 0$).

9. ONE-WAY ANOVA [G — THE NUMERICAL QUESTION]

$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $i \leq r$, $j \leq n_i$, $n = \sum n_i$; $\rho(\mathbf{X}) = r$ (not $r+1$). **Identifiability constraint** $\sum_i w_i \alpha_i = 0$, $\sum w_i = 1$; natural choice $w_i = n_i / n$ i.e.

$$\sum_i n_i \alpha_i = 0$$

(The constraint is a labelling convention: SSE, F , fitted values do NOT depend on it.) **Estimable:** μ_i ; $\alpha_i - \alpha_{i'}$; any contrast $\sum c_i \alpha_i$ with $\sum c_i = 0$. **Not:** μ or α_i alone.

$$\hat{\mu}_i = \bar{Y}_{i.}, \quad \hat{\mu} = \bar{Y}_{..}, \quad \hat{\alpha}_i = \bar{Y}_{i.} - \bar{Y}_{..}$$

COMPUTATIONAL RECIPE. $T_i = \sum_j Y_{ij}$, $G = \sum_i T_i$, CF = G^2 / n :

$$\text{SSTO} = \sum \sum Y_{ij}^2 - \text{CF}$$

$$\text{SSTR} = \sum_i T_i^2 / n_i - \text{CF}$$

$$\text{SSE} = \text{SSTO} - \text{SSTR}$$

Source	d.f.	SS	MS	$\mathbb{E}(\text{MS})$
Treatments	$r - 1$	SSTR	MSTR	$\sigma^2 + \frac{\sum n_i (\mu_i - \mu)^2}{r - 1}$
Error	$n - r$	SSE	MSE	σ^2
Total	$n - 1$	SSTO		

$H_0 : \alpha_1 = \dots = \alpha_r = 0$:

$$F = \text{MSTR} / \text{MSE} \sim F_{r-1, n-r}$$

reject if $F > F_{r-1, n-r}^{1-\alpha}$. (Under H_1 : non-central with $\lambda = \sum n_i \alpha_i^2 / \sigma^2$.) $\text{MSE} = \sum_i (n_i - 1) s_i^2 / (n - r)$ (pooled).

INTERVALS ($t^* = t_{n-r}^{1-\alpha/2}$, all use MSE).

$$\mu_i : \bar{Y}_{i.} \pm t^* \sqrt{\frac{\text{MSE}}{n_i}} \quad \mu_i - \mu_{i'} : \bar{Y}_{i.} - \bar{Y}_{i'} \pm t^* \sqrt{\text{MSE} \left(\frac{1}{n_i} + \frac{1}{n_{i'}} \right)}$$

Contrast $\theta = \sum c_i \mu_i$, $\sum c_i = 0$: BLUE $\hat{\theta} = \sum c_i \bar{Y}_{i.}$, $\text{Var} = \sigma^2 \sum \frac{c_i^2}{n_i}$,

$$\sum_i c_i \bar{Y}_{i.} \pm t^* \sqrt{\text{MSE} \sum_i c_i^2 / n_i}$$

CALCULATOR CHECKS. $\sum T_i = G$; $\text{SSTR} + \text{SSE} = \text{SSTO}$; $(r-1) + (n-r) = n-1$; $\sum n_i \alpha_i = 0$; all SS ≥ 0 ; keep MSE to 4 d.p. $r = 2 \Rightarrow F = T^2$ (**two-sample t , pooled over all groups**).

10. QUANTILES (insurance)

ν	.95	t_ν	.975	.995	$F_{d_1, \nu}^{0.95}$	$d_1=1$	2	3	4	5
4	2.132	2.776	4.604	7.71	6.94	6.59	6.39	6.26		
5	2.015	2.571	4.032	6.61	5.79	5.41	5.19	5.05		
6	1.943	2.447	3.707	5.99	5.14	4.76	4.53	4.39		
8	1.860	2.306	3.355	5.32	4.46	4.07	3.84	3.69		
9	1.833	2.262	3.250	5.12	4.26	3.86	3.63	3.48		
10	1.812	2.228	3.169	4.96	4.10	3.71	3.48	3.33		
12	1.782	2.179	3.055	4.75	3.89	3.49	3.26	3.11		
15	1.753	2.131	2.947	4.54	3.68	3.29	3.06	2.90		
16	1.746	2.120	2.921	4.49	3.63	3.24	3.01	2.85		
18	1.734	2.101	2.878	4.41	3.55	3.16	2.93	2.77		
20	1.725	2.086	2.845	4.35	3.49	3.10	2.87	2.71		
21	1.721	2.080	2.831	4.32	3.47	3.07	2.84	2.68		
24	1.711	2.064	2.797	4.26	3.40	3.01	2.78	2.62		
27	1.703	2.052	2.771	4.21	3.35	2.96	2.73	2.57		
30	1.697	2.042	2.750	4.17	3.32	2.92	2.69	2.53		
40	1.684	2.021	2.704	4.08	3.23	2.84	2.61	2.45		
60	1.671	2.000	2.660	4.00	3.15	2.76	2.53	2.37		

$z_{.95} = 1.645$, $z_{.975} = 1.960$, $z_{.995} = 2.576$. $t_\nu^2 = F_{1, \nu}$.

11. DECISION TREE

- “Is $a^T \boldsymbol{\beta}$ estimable?” $\rightarrow$ find $\mathcal{N}(\mathbf{X})$, check $a^T \mathbf{z} = 0$. **First compute $\rho(\mathbf{X})$ — it may be full rank.**
- “Find the BLUE” $\rightarrow$ Gauss–Markov: $a^T \hat{\boldsymbol{\beta}}$; get $\hat{\boldsymbol{\beta}}$ from cell means. Variance = $\sigma^2 a^T (\mathbf{X}^T \mathbf{X})^{-1} a$.
- “Propose a test for H_0 ” $\rightarrow$ write it as $L^T \boldsymbol{\beta} = \theta_0$, quote $W/r \sim F_{r, n-p}$ or $\frac{(\text{SSE}_{red} - \text{SSE}_{full}) / r}{\text{MSE}}$. One constraint $\Rightarrow$ use t .
- “Show $\dots \sim \chi^2$ ” $\rightarrow$ show the matrix is symmetric idempotent, quote $\rho = \text{tr}$.
- “Show independence” $\rightarrow A \Sigma B^T = 0$ (linear/linear) or $B \Sigma A = 0$ (linear/quadratic), or Fisher–Cochran d.f. sum.
- “Show G is a g-inverse” $\rightarrow$ compute AGA and use $\rho(A) = \rho(A_{11})$ to write the dependent rows.
- Numerical $\rightarrow$ totals table, CF, SS by shortcut, ANOVA table, then contrasts.